

SARA AMIRI

Chemical & Biotechnology Engineering Student | Seeking Internship from February 2027

Copenhagen, Denmark | +45 60 90 53 55 | Amyri9080@gmail.com



PROFILE

Chemical and Biotechnology Engineering student at the Technical University of Denmark (DTU) with hands-on experience in analytical chemistry, food analysis, biochemical engineering and process design. Starting my Bachelor's thesis in autumn 2026 on nitrogen-to-protein conversion factors for the valorization of crab processing side-streams, involving Dumas protein analysis, amino acid profiling and statistical data analysis. Seeking an internship from February 2027, where I can contribute strong analytical and laboratory skills within biotechnology, food, pharmaceutical or QC/QA environments.

TECHNICAL & LABORATORY SKILLS

- Analytical & food chemistry:**Dumas protein analysis, amino acid profiling, GC-FID, FAME preparation, lipid extraction, fatty acid profiling, spectrophotometry, chromatography, mass balances, water activity and colour measurement
- Data & statistics:** Hypothesis testing, confidence intervals, ANOVA, regression, data visualisation and critical evaluation of analytical data
- Reaction engineering:**Batch, CSTR, PFR and PBR reactor design, kinetics, conversion, RTD and non-isothermal energy balances
- Process engineering:**Heat exchangers, evaporation, absorption/stripping, fermentation, mass and energy balances and process optimisation
- Quality:** GMP, quality management, ISO principles, analytical quality control and documentation
- Software:** Python, MATLAB, Maple, Excel, Word and PowerPoint

SELECTED PROJECT EXPERIENCE

Bachelor's Thesis - DTU Food Autumn 2026 – starting
Determination of Nitrogen-to-Protein Conversion Factors for the Valorization of Crab Processing Side-Streams — Food analysis and analytical chemistry

- Performing total nitrogen determination using the Dumas method on different crab fractions
- Conducting amino acid profiling, distinguishing between essential and non-essential amino acids
- Performing mass balance calculations and statistical evaluation of nitrogen and amino acid profiles
- Developing matrix-specific nitrogen-to-protein conversion factors in collaboration with Avedøreværket
- Supporting sustainable valorization of crab processing side-streams

Chemical Analysis of Plant-Based Food Products - DTU Food
Food analysis and analytical chemistry

- Performed lipid extraction, FAME preparation and GC-FID analysis of plant-based spreads
- Evaluated fatty acid composition (SFA, MUFA, PUFA) and compared results with product composition
- Applied analytical data criticism, assessed sources of error, method quality and reproducibility

Heat Exchanger Design and Sterilisation of Fermentation Substrate

- Designed and optimised a heat exchanger using thermodynamic and fluid-dynamic principles
- Worked with sterilisation of fermentation substrates and evaluation of process efficiency

Bioethanol and Cellulase Production Biotechnology and process design

- Worked with fermentation, oxygen transfer (kLa/DO), process conditions and biochemical process optimisation
- Applied process engineering principles to bioethanol and cellulase production

Cumene Manufacturing Process Chemical process design

- Analysed and optimised a cumene production process using heat exchange, flash separation and process calculations
- Applied mass balances, process design and economic evaluation concepts

Design of a Multiple-Effect Evaporation System for Sugar Production

- Designed an evaporation system for sugar production with focus on energy efficiency and process optimisation

EDUCATION

Bachelor of Engineering in Chemical and Biotechnology Engineering Expected graduation: 2027
Technical University of Denmark (DTU), Copenhagen

Relevant coursework: Statistics, Chemical Reaction Engineering, Materials Science, Chemical Food Analysis, Analytical Chemistry, Chemical and Biochemical Process Engineering, Unit Operations, Biotechnology and Process Design, Mathematical Modelling, Physical Chemistry, Biological Chemistry, Process Design

CERTIFICATES & ADDITIONAL COURSES

- Quality Management and Good Manufacturing Practice (GMP) — quality assurance, ISO principles and regulated industries
- Project Management — planning, progress monitoring, risk assessment, stakeholder analysis, budgeting and project leadership

LANGUAGES

Danish | English | Persian